EXPERIMENT NO : 8

CONDUCT Z TEST FOR THE DATA GIVEN USING PYTHON
OBJECTIVE

AIM:

To Conduct z test for the data given using python objective.

CODE:

```
import numpy as np
import scipy.stats as stats

# Step 1: Input sample data as a single string instead of a multiline array data_string
= "152 148 151 149 147 153 150 148 152 149 151 150 149 152 151 148 150 152
149 150 148 153 151 150 149 152 148 151 150 153"
sample_data = np.fromstring(data_string, sep=" ")

# Step 2: Population mean under Null Hypothesis (H0: mu = 150)
population_mean = 150

# Step 3: Calculate sample statistics
sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data,
ddof=1) n = len(sample_data)

# Step 4: Calculate T-statistic and P-value t_statistic, p_value =
stats.ttest_1samp(sample_data, population_mean)

# Step 5: Print results print(f"Sample
Mean: {sample_mean:.2f}")
print(f"Sample Std Dev:
{sample_std:.4f}") print(f"Degrees of
Freedom: {n - 1}") print(f"T-Statistic:
{t_statistic:.4f}") print(f"P-Value:
{p_value:.4f}")

# Step 6: Decision
rule alpha = 0.05 if
p_value < alpha:
```

```
    print("Decision: Reject the null hypothesis (Significant
difference).") else:      print(
        "Decision: Fail to reject the null hypothesis (No significant difference)."
    )
```

OUTPUT:

```
            Sample Mean: 150.20
            Sample Std Dev: 1.7100
Degrees of Freedom: 29
            T-Statistic: 0.6406
             P-Value: 0.5268
            Decision: Fail to reject the null hypothesis (No significant difference)
```

RESULT:

Thus, the code was analyzed successfully and the output was verified.